# Refinement versus Refresh: Compute Allocation for Closed-Loop Robot Policies

**Alba María Téllez Fernández**
MAAT Institute
alba.tellez@maatinstitute.com

## Abstract

How should a chunked robot policy spend inference compute: refine its current action chunk, or replan from a newer observation? We present a case-study protocol for answering this question within a frozen policy. The protocol treats denoising depth and executed prefix as separate controls, calibrates complete generation cost, separates configuration discovery from held-out confirmation, and evaluates closed-loop success together with realized episode expenditure. Applied to SmolVLA in synchronous LIBERO, the study shows why both controls must be considered jointly. At a fixed refresh schedule, fewer denoising steps improved success while reducing arithmetic. At low denoising, a longer execution prefix substantially reduced expenditure while producing similar observed success, although equivalence was not established. Most generation arithmetic was independent of denoising, so reducing refinement alone could not deliver proportional savings. A fitted response model offered no demonstrated selection advantage over a simple cost-aware rule. These findings are checkpoint-specific. The broader contribution is a reproducible procedure for testing refinement–refresh allocations without treating arithmetic efficiency, latency, or apparent quality preservation as interchangeable.

## 1 Introduction

A robot policy can reduce inference work in two ways: spend less computation refining each action prediction, or execute more actions before requesting a new prediction. These interventions affect behavior differently. Reduced refinement changes the prediction itself; longer execution prefixes delay observation-conditioned revision and expose later positions of the predicted action sequence. Their relative value should therefore be evaluated through closed-loop outcomes under a common resource constraint.

We ask: *given a fixed inference budget, how should a robot divide computation between prediction refinement and observation-conditioned refresh?* The cost of a fresh prediction makes this question nontrivial. In our frozen SmolVLA implementation [1], work independent of denoising accounts for 82.1% of native counted arithmetic. Reducing refinement therefore leaves a substantial cost per generation: sufficiently tight budgets require less frequent refresh even at low denoising counts. This cost boundary determines what is feasible, but not which feasible setting controls the robot best.

We operationalize this allocation question through two within-checkpoint comparisons. The primary comparison varies denoising at a fixed execution prefix to evaluate whether additional refinement improves success. A complementary, descriptive comparison varies the execution prefix at fixed denoising to examine whether more frequent refresh justifies its additional expenditure. We evaluate twelve configurations in a paired 480-attempt discovery study, then seven configurations in 1,120 confirmation attempts on eight tasks excluded from selection. Three frozen comparisons evaluate

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refinement. The later-amended prefix comparisons examine refresh; they neither isolate observation freshness from action-position effects nor establish an optimal allocation rule.

At the primary two-action prefix, two denoising steps yield 139/160 successes versus 122/160 for five steps, with lower reference arithmetic. At two denoising steps, however, a five-action prefix yields 138/160 successes at 60.7% less arithmetic per attempt than the two-action prefix; their quality difference remains unresolved. A simple cost-aware rule also matches the fitted response model's choices at all three ceilings, with no demonstrated selection advantage from fitting.

The contributions are:

- a reusable refinement–refresh evaluation procedure combining full-generation calibration, explicit compute ceilings, frozen comparisons, and closed-loop outcome measurement;
- a paired twelve-configuration study reporting both nominal compute rate and realized episode expenditure; and
- a completed task-transfer evaluation separating frozen refinement tests from descriptive refresh comparisons, with their uncertainty and selection provenance retained.

## 2 Related Work

Resource-allocation studies motivate evaluating configurations across explicit budgets. Chinchilla studies model-size and training-data allocation [2]; embodied scaling studies consider world modeling, behavior cloning, and closed-loop forecasting and planning [3, 4]. Our comparison instead varies deployment refinement and execution prefix within a fixed policy. Work distinguishing FLOP-optimal from elapsed- or monetary-cost-optimal allocation [5] also motivates retaining separate resource coordinates.

VLA-Perf characterizes inference performance across denoising, chunk size, asynchronous execution, and hardware/runtime choices [6]. SmolVLA studies asynchronous inference and robot behavior [1]. Here predicted chunk size remains fixed, while executed prefix changes; task success is measured alongside computation. RoboMonkey studies test-time sampling and action verification [7], an adjacent allocation axis that would require counting candidate generation and verification costs.

Adaptive VLA scheduling is directly related. DVAC uses denoising-derived action variance to choose an execution prefix before replanning [8]. ElegantVLA schedules vision–language recomputation or reuse together with action denoising modes [9]. Our study instead compares fixed configurations under explicit arithmetic ceilings and freezes separate-task denoising comparisons. It neither introduces an adaptive scheduler nor establishes an advantage over these methods; their reported speedups have different policies, hardware, and measurement boundaries.

## 3 Refinement–Refresh Allocation

### 3.1 Compute rate and episode expenditure

Let $K$ be the predicted action-chunk length and $h < K$ the execution prefix. At each policy query, the policy predicts $K$ actions, executes only the first $h$, discards the remaining $K - h$, and then generates a new chunk from a new observation. Thus $h$ controls replanning frequency without changing the predicted chunk length. In all experiments, $K = 50$. We reserve $T_e$ for the number of actions actually executed in attempt $e$. Let $d$ be the denoising count and $F_{\text{gen}}(d)$ the reference-calibrated neural arithmetic for a full action-chunk generation at denoising setting $d$. For complete execution prefixes, the nominal compute rate per simulated control step is

$$b(h, d) = \frac{F_{\text{gen}}(d)}{h}.$$

For a candidate set $\mathcal{C}$ and budget ceiling $B$, the allocation target is

$$(h^\star, d^\star)(B) \in \arg \max_{(h,d)\in\mathcal{C}:\, b(h,d)\le B} p_{\text{success}}(h, d). \tag{1}$$

A configuration need not exhaust the ceiling. This is a deployment compute-rate constraint for a fixed checkpoint, rather than a training-plus-deployment objective.

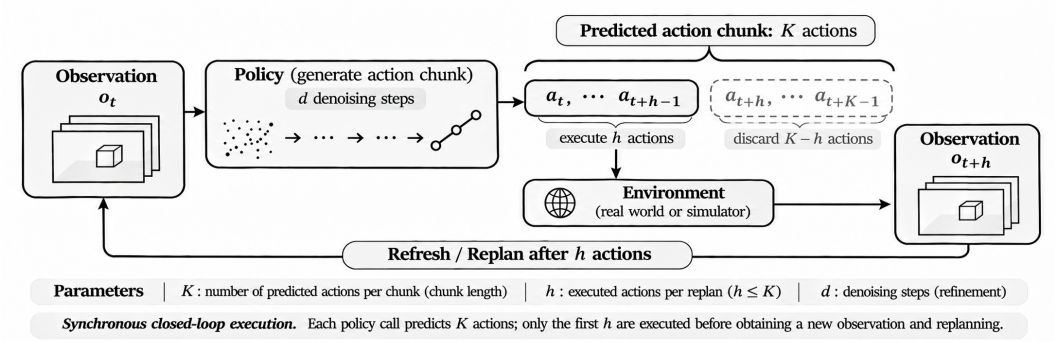



Figure 1: Synchronous chunked closed-loop execution. A policy call predicts $K$ actions after $d$ denoising steps; only the first $h$ actions are executed, with the remaining $K - h$ actions discarded before the next observation and replanning step. In the experiments, $K = 50$.

Figure 1 summarizes the execution loop. Each policy call predicts a fixed-length chunk, executes only the first $h$ actions, discards the remainder, and refreshes from the resulting observation. The denoising count $d$ controls refinement inside the policy call, while $h$ controls how frequently the policy receives a new observation.

We distinguish this rate from total arithmetic per attempted episode. If attempt $e$ executes $T_e$ actions, its generated-chunk expenditure is

$$C_e = \lceil T_e/h \rceil \, F_{\text{gen}}(d). \tag{2}$$

The final generated chunk is charged even if only part of its execution prefix is used. All-attempt means include unsuccessful episodes. Different trajectories and termination times can therefore change expenditure at the same nominal rate. FLOPs, host elapsed time, and CUDA-stream time retain separate units and measurement boundaries.

### 3.2 The fixed cost of a fresh prediction

Complete-generation profiling counts matrix products, convolutions, and attention across vision, language prefill, and the action expert, using two FLOPs per multiply-add. Embedding lookups, normalization, activations, elementwise arithmetic, and external preprocessing are excluded. We therefore refer to *dense neural arithmetic*, not an exact count of every scalar operation. Calibration uses one real development observation and fixed tensor shapes; its protocol is detailed in Appendix A.

For the reference input, the measured arithmetic satisfies, to rounding,

$$F_{\text{gen}}(d) \simeq 527.108 + 11.457d \quad \text{GFLOPs}$$

at all six calibrated denoising settings. Native generation at $d = 10$ costs 641.675 GFLOPs, of which 82.1% is independent of $d$. Halving denoising from ten to five saves 8.9% of counted generation work; reducing it to two saves 14.3%. Equivalently, relative to native deployment $(h, d) = (1, 10)$,

$$\frac{b(h,d)}{b_{\text{native}}} \simeq \frac{0.821 + 0.179(d/10)}{h}.$$

We prespecify compute-rate ceilings of $0.25$, $0.50$, and $1.00$ times native. Eligibility uses measured per-$d$ values, not the rounded affine expression. For example, $(1, 2)$ costs 0.857 times native rate and is ineligible at half-native budget, whereas $(2, 2)$ costs 0.429 times native. The denoising-independent work alone excludes $h = 1$ at half-native budget and $h \leq 2$ at quarter-native budget. A tighter budget can therefore force a longer execution prefix before any quality-based choice is made. This is an implementation-specific arithmetic floor under the declared counting boundary, not a universal VLA scaling law.

Calibration uses a fixed input shape; a separate language-length audit leaves eligibility unchanged under the original absolute ceilings (Appendix A.3). Reference arithmetic remains distinct from task-shaped arithmetic and elapsed time. The cost model defines feasible schedules; the following experiments ask which feasible expenditures help closed-loop control.

# 4 Controlled Study and Frozen Comparisons

## 4.1 Discovery and confirmation protocol

We evaluate the public `HuggingFaceVLA/smolvla_libero` checkpoint on LIBERO [10]. The predicted chunk length remains fixed at $K = 50$; native execution uses $(h, d) = (1, 10)$. We hold weights, sensors, preprocessing, action normalization, control period, and task-specific episode horizons fixed.

Discovery crosses $h \in \{1, 2, 5\}$ and $d \in \{2, 5, 10, 20\}$ on Spatial tasks 5–8, five initial states per task, and two policy seeds. Each of the 40 paired task–state–seed identities receives all twelve configurations, giving 480 attempts. Tasks 5 and 6 overlap the pilot; tasks 7 and 8 were selected before observing their outcomes in this study. Condition order is randomized within each paired identity.

Discovery is used to freeze three budget-specific denoising comparisons before confirmation. Their selected settings and comparators, together with native, give seven configurations. Confirmation evaluates all seven on Goal and Object tasks 5–8, ten states per task, and two seeds: 160 attempts per configuration, totaling 1,120. These eight tasks are excluded from configuration selection; prior exposure during unrelated development is not assessed. The sample allocation prioritizes task and state coverage; no quality-preservation margin is prespecified. Appendix A.5 distinguishes the original tests from the later-amended descriptive comparisons.

Initial simulator states and observations match across conditions. Policy noise is keyed by task, state, seed, and simulation tick, preserving a common noise key whenever conditions generate at the same tick. Solver timesteps and a constant-field integration check verify that every $d$ traverses the full interval from 1 to 0; fewer steps change the discretization, not the integration endpoint. Recorded initial conditions, generated and executed actions, and outcomes agree across the evaluation records. LIBERO's sparse reward and success indicator derive from the same task-goal evaluation, so their agreement checks record consistency rather than supplying an independent outcome measure. The recorded outcome path is traced to the benchmark's task-specific goal conditions. These checks establish strong internal consistency rather than an independent simulator replay (Appendix A.5).

Success averages both seeds within each state, then states and tasks with equal task weights. Descriptive 95% intervals use 20,000 bootstrap draws of states within the fixed tasks, retaining paired seeds and configurations. They condition on four discovery or eight confirmation tasks; secondary intervals are unadjusted. Arithmetic includes every generated chunk, including failures and partly consumed final prefixes. Timing measurements and their monitoring limits are reported separately in Appendix A.1.

## 4.2 Discovery: quality does not increase monotonically with refinement

Native $(1, 10)$ succeeds in 25/40 attempts. The highest observed count is 28/40 for $(1, 5)$; $(1, 2)$ and $(2, 2)$ each succeed in 26/40. At $h = 5$, successes are 21, 18, 16, and 14 out of 40 as $d$ increases from 2 to 20. Figure 2(a) shows every configuration. These point estimates do not justify imposing monotonic improvement with denoising. The small paired sample also does not certify equal quality for configurations with similar observed counts.

The resource coordinate also matters: nominal rate differs from realized episode expenditure, and arithmetic savings differ from elapsed savings. Figure 2(b) illustrates how substituting a host-time constraint changes feasible refresh choices. This sensitivity analysis does not change the frozen arithmetic-based experiment. Full task and resource tables appear in Appendix A.

## 4.3 Frozen tests and the role of selection

A response model fitted only to discovery selects $(5, 2)$, $(2, 2)$, and $(1, 2)$ at the quarter-, half-, and full-native ceilings. Their frozen comparators are $(5, 5)$, $(2, 5)$, and $(1, 5)$, respectively; the half-native comparison is primary. All selected settings are existing discovery grid points, so confirmation tests task transfer rather than extrapolation to an untested configuration. Each pair changes denoising at a fixed prefix. These are comparisons under common ceilings, not equal-rate tests of how to divide compute between refinement and refresh.

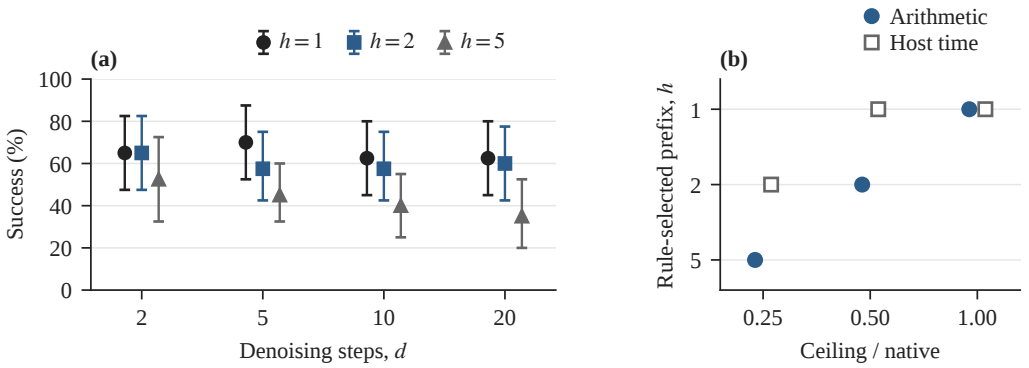



Figure 2: Quality and feasibility are distinct. (a) Discovery success, with 40 attempts per configuration. Colors and shapes identify executed prefix $h$; bars give descriptive, task-conditional 95% paired state-bootstrap intervals. The $d$ axis is categorical; horizontal offsets only separate overlapping bars. (b) A minimum-denoising, shortest-feasible-prefix rule selects different prefixes under separately normalized arithmetic and fixed-input host-time ceilings. The $h$ axis shows the selected prefix categories, not elapsed time. These are feasibility-dependent choices, not established quality optima; timing details are in Appendix A.2.

At the primary ceiling, $(2, 2)$ has the highest eligible discovery success. Its comparator $(2, 5)$ ties with $(2, 10)$ for the highest count among distinct eligible alternatives and wins the lower-cost tie-break. A simple rule choosing minimum denoising and then the shortest feasible prefix makes the same choices as the fitted model at all three ceilings. This baseline was documented after collection began and is interpreted descriptively. Transfer outcomes cannot demonstrate a selection advantage between rules that make identical choices. Model fitting, the full candidate set, and selector diagnostics are retained in Appendix B.

## 5 Refinement and Refresh on Transfer Tasks

### 5.1 Complete confirmation overview

All 1,120 originally planned confirmation attempts are complete. Table 1 reports the seven configurations on the same paired identities. The audit excludes two repeated completed records under the documented precedence rule; continuation completes the original design, not an independent replication (Appendix A). We first report the frozen denoising tests, then the amended prefix comparisons.

Table 1: Confirmation overview: 160 attempts per configuration on eight tasks. $K = 50$ is the fixed predicted chunk length, $h$ is the executed prefix, and $d$ is the denoising count. Nominal rate/native is $F_{\text{gen}}(d)/(hF_{\text{gen}}(10))$. All-attempt arithmetic uses the length-20 reference calibration and charges every generation. The primary pair is $(2, 2)$ versus $(2, 5)$.

| Allocation | | Success | | Nominal rate | Mean per attempt | |
|---|---|---|---|---|---|---|
| $h$ | $d$ | Count | (%) | / native | TFLOPs | Steps |
| 1 | 2 | 137/160 | 85.6 | 0.857 | 83.3 | 151.4 |
| 1 | 5 | 136/160 | 85.0 | 0.911 | 89.1 | 152.5 |
| 1 | 10 | 135/160 | 84.4 | 1.000 | 98.4 | 153.4 |
| 2 | 2 | 139/160 | 86.9 | 0.429 | 41.1 | 149.0 |
| 2 | 5 | 122/160 | 76.3 | 0.455 | 48.7 | 166.3 |
| 5 | 2 | 138/160 | 86.3 | 0.171 | 16.1 | 144.9 |
| 5 | 5 | 133/160 | 83.1 | 0.182 | 18.5 | 156.2 |

### 5.2 Refinement: the frozen denoising comparisons

At the primary two-action prefix, two denoising steps yield 139/160 successes versus 122/160 for five steps. The paired difference is $+10.6$ percentage points, with descriptive 95% interval $[+4.4, +16.9]$, conditional on the eight tasks. The selected setting also uses 5.9% less nominal reference arithmetic per control step: 275.0 versus 292.2 GFLOPs. Thus reducing refinement improves observed success while reducing counted work at this fixed prefix.

Mean reference arithmetic per attempt falls from 48.7 to 41.1 TFLOPs (15.6%). This larger reduction includes changed trajectories and termination: mean episode length is 149.0 versus 166.3 control steps. All failures reach the task horizon (21 versus 38); the remaining attempts terminate successfully. The primary direction is positive on seven tasks and tied on one, with 24 paired wins and seven losses. Appendix A.4 gives task sensitivity and expenditure intervals. These measurements do not establish a controlled physical-latency improvement. Figure 3 places the primary effect alongside all other frozen and amended comparisons.

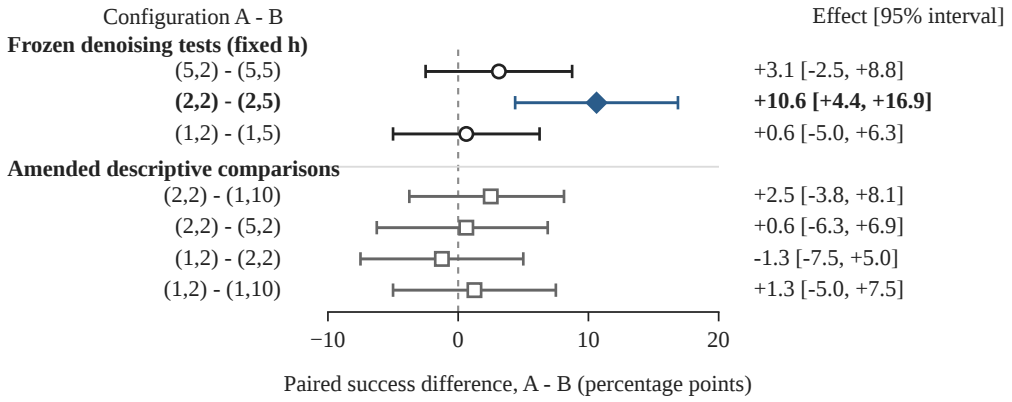



Figure 3: Confirmation effects: 160 paired attempts per setting on eight tasks. Rows compare $(h, d)$ as A minus B; bars are task-conditional, descriptive 95% paired state-bootstrap intervals (secondary intervals unadjusted). The filled diamond marks the prespecified primary test. All three frozen tests and all four amended contrasts appear in their documented order; the latter were not fully outcome-blind. Intervals crossing zero do not establish equivalence. Predicted and observed gaps are tabulated in Tables 11 and 9.

The other two frozen tests have positive point estimates but intervals spanning zero (Figure 3). At $h = 5$, two versus five denoising steps give 138/160 versus 133/160 successes; at $h = 1$, they give 137/160 versus 136/160. The largest observed benefit is at the primary prefix. The primary gap exceeds its frozen prediction, while matching the signs of three point estimates does not establish calibrated effect-size prediction. These results do not demonstrate a uniform denoising benefit across prefixes.

### 5.3 Refresh: descriptive prefix comparisons

We now hold denoising fixed and compare expenditures on different refresh schedules. These contrasts were documented after collection began, when individual outcomes were visible; they are not originally prespecified or fully outcome-blind (Appendix A.5).

At $d = 2$, $(2, 2)$ exceeds $(5, 2)$ by just one success: 139/160 versus 138/160, a paired difference of $+0.6$ percentage points $[-6.3, +6.9]$. Yet $(5, 2)$ uses 60% less nominal arithmetic per control step and 60.7% less reference arithmetic per attempt: 16.1 versus 41.1 TFLOPs. The interval permits meaningful differences in either direction; similar counts do not certify equivalent quality or justify a non-inferiority claim.

Refreshing every action with $(1, 2)$ instead of every two actions gives two fewer successes, a difference of $-1.3$ points $[-7.5, +5.0]$, at 83.3 versus 41.1 TFLOPs per attempt. More frequent refresh has no demonstrated quality benefit in these contrasts. Native-relative comparisons, which also change denoising, are reported separately in Appendix A.5.

The discovery point estimates favor shorter prefixes at $d = 2$ (Section 4.2), but their quality advantage is not established on transfer. The task populations differ, so this comparison does not identify a causal task effect. The cost model explains feasibility, but does not determine the quality ranking on new tasks.

## 6 Discussion and Conclusion

This study develops a procedure for evaluating how chunked robot policies allocate inference compute between refining an action prediction and refreshing it from a new observation. The procedure measures the complete cost of a policy call, separates configuration discovery from held-out confirmation, freezes the comparisons before evaluation, and reports closed-loop success together with episode expenditure. This prevents a cheap configuration from being treated as useful without behavioral evidence, or a successful configuration from being discussed without its compute cost.

The SmolVLA case study demonstrates why this joint evaluation is necessary. Additional refinement did not consistently produce better closed-loop behavior, and reducing refinement alone delivered limited savings because most generation work was independent of denoising. Changing the execution prefix affected cost more strongly, but similar observed success was insufficient to establish equivalent quality. The fitted response model also did not improve configuration choices over a simple cost-aware rule. The conclusion is methodological: refinement and refresh should be evaluated together within each model, under explicit counting boundaries and prespecified comparisons. The case study does not establish a universal allocation.

**Interpretation and limits.** The two deployment controls are not isolated causal interventions on prediction quality and observation freshness. Changing denoising changes numerical integration; changing prefix changes both observation age and executed action positions. The mechanism behind the benefit of coarser sampling remains open. The study covers one checkpoint and twelve fixed tasks in simulation, with only four discovery task-level validation folds. Intervals are task-conditional, secondary contrasts are unadjusted, and no quality-preservation margin was prespecified. Two steps is the minimum denoising tested; $d = 1$ is untested. The arithmetic boundary excludes several operation classes; language lengths are audited but image shape is fixed. Sampled GPU checks have coverage gaps, and simulated control steps do not establish real-time responsiveness. The recorded outcomes trace to LIBERO's task-goal evaluation, and one native-setting parity episode reproduces the reference evaluator's initial state, action sequence, episode length, and outcome. However, the confirmation actions were not replayed in a separate, version-matched simulator. Outcome validity is therefore strongly supported internally but not fully independently established.

**Next steps.** A controlled stale-input diagnostic is needed to separate observation freshness from the action-position effects introduced by changing the executed prefix. A predeclared quality-preservation test should evaluate the cheaper $(5, 2)$ setting, and independent outcome replay should test the retained primary trajectories in a separate, version-matched simulator. The same study design can then be applied within other policies and execution stacks: define the refinement and refresh controls, calibrate complete inference cost, freeze the comparisons, and evaluate success and expenditure jointly. Replication across checkpoints, policy architectures, benchmark families, and asynchronous or physical-robot execution is required to determine when the fixed cost of a fresh prediction is worth paying and how much refinement that prediction needs.

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# A Detailed Outcomes and Resource Measurements

The checkpoint is `HuggingFaceVLA/smolvla_libero`, revision `6721902bc4d61e50a3bfdb11dfb4cb626f05d102`. Predicted chunk size is 50; native execution uses $h = 1, d = 10$. Discovery uses Spatial tasks 5–8, initial states 10–14, and policy seeds 2000 and 2001. The 20 task–state blocks contain both seeds and all 12 configurations. Confirmation uses Goal and Object tasks 5–8, states 20–29, and policy seeds 3000 and 3001. Its 80 task–state blocks each contain both seeds and all seven configurations. Discovery bootstrap resampling uses 20,000 draws with analysis seed 20260913.

Table 2: Discovery successes on Spatial tasks 5–8. Each task contributes five states and two seeds per configuration. Rate/native is $F_{\text{gen}}(d)/(hF_{\text{gen}}(10))$, not realized episode cost. The best observed count is 28/40 at $(1, 5)$; no monotonic denoising trend is assumed.

| Allocation | | Spatial task successes (each /10) | | | | | Nominal rate |
|---|---|---|---|---|---|---|---|
| $h$ | $d$ | Task 5 | Task 6 | Task 7 | Task 8 | Total | / native |
| 1 | 2 | 5 | 7 | 4 | 10 | 26/40 | 0.857 |
| 1 | 5 | 5 | 8 | 6 | 9 | 28/40 | 0.911 |
| 1 | 10 | 5 | 7 | 6 | 7 | 25/40 | 1.000 |
| 1 | 20 | 5 | 6 | 6 | 8 | 25/40 | 1.179 |
| 2 | 2 | 5 | 7 | 4 | 10 | 26/40 | 0.429 |
| 2 | 5 | 4 | 6 | 4 | 9 | 23/40 | 0.455 |
| 2 | 10 | 4 | 6 | 4 | 9 | 23/40 | 0.500 |
| 2 | 20 | 4 | 7 | 4 | 9 | 24/40 | 0.589 |
| 5 | 2 | 6 | 5 | 4 | 6 | 21/40 | 0.171 |
| 5 | 5 | 2 | 5 | 5 | 6 | 18/40 | 0.182 |
| 5 | 10 | 2 | 4 | 5 | 5 | 16/40 | 0.200 |
| 5 | 20 | 2 | 3 | 4 | 5 | 14/40 | 0.236 |

Calibration uses one real observation from Spatial task 0, initial state 0, and common explicit noise. For each $d \in \{2, 3, 5, 8, 10, 20\}$, profiling records solver timesteps and dense operation counts, separately from elapsed timing. Each setting receives 20 warm-up generations, followed by three randomized timing blocks of 20 measured calls per setting. The rollout evaluator also performs 20 warm-up generations once per configuration.

Host generation timing brackets prediction and completion synchronization; CUDA events measure the current-stream prediction interval. The outer rollout clock additionally includes observation processing, action handling, simulation/rendering, and record writing. Model loading, environment creation, reset, and warm-up are measured separately. Process sampling occurs approximately every five seconds; the largest recorded gap during discovery is below 5.03 seconds. Display processes remain on the measurement GPU.

Calibration requires approximately 291 seconds and discovery 31,245 seconds (8.68 hours) of host elapsed time, including setup and bookkeeping. This search expenditure is additional to subsequent deployments. The experiment does not estimate an amortization threshold or a training–deployment cost crossover.

All measurements use an RTX 4060 Ti. Five-second process samples show no unrelated CUDA compute process during calibration, discovery, or the initially scheduled transfer phase; display processes remain. Process samples were not retained for priority completion or the beginning of continuation. Later continuation monitoring also shows no unrelated sampled CUDA compute process, but these coverage gaps make aggregate transfer rollout times descriptive. Host and CUDA-stream generation intervals are measured separately from the outer synchronous rollout interval. Warm-up is excluded from these rollout measurements.

### A.1 Fixed-input generation timing

The calibration stores full-generation intervals and global operator arithmetic counts, not separate vision, language-prefill, or action-expert timings and arithmetic totals. These records cannot attribute elapsed sensitivity to a particular component, memory traffic, omitted operation classes, synchroniza-

Table 3: Discovery resource measurements. Generated-chunk totals use the declared dense-arithmetic boundary and charge partially executed final prefixes. Episode means include unsuccessful attempts. $(2, 2)$ has 26/40 observed successes at 42.9% of native arithmetic rate.

| Allocation | | Nominal / step | Mean per attempt | | Success |
|---|---|---|---|---|---|
| $h$ | $d$ | GFLOPs | TFLOPs | Rollout (s) | Count |
| 1 | 2 | 550.0 | 90.6 | 39.5 | 26/40 |
| 1 | 5 | 584.4 | 91.7 | 61.8 | 28/40 |
| 1 | 10 | 641.7 | 109.6 | 111.9 | 25/40 |
| 1 | 20 | 756.2 | 129.1 | 199.8 | 25/40 |
| 2 | 2 | 275.0 | 45.0 | 21.8 | 26/40 |
| 2 | 5 | 292.2 | 53.4 | 38.5 | 23/40 |
| 2 | 10 | 320.8 | 57.8 | 61.6 | 23/40 |
| 2 | 20 | 378.1 | 67.8 | 107.8 | 24/40 |
| 5 | 2 | 110.0 | 20.5 | 12.7 | 21/40 |
| 5 | 5 | 116.9 | 23.8 | 20.2 | 18/40 |
| 5 | 10 | 128.3 | 27.2 | 32.0 | 16/40 |
| 5 | 20 | 151.2 | 33.1 | 55.6 | 14/40 |

Table 4: Fixed-input full-generation calibration. Each setting has 60 calls in three randomized blocks. Range gives the minimum and maximum host block means, not a confidence interval. Host time falls more steeply with $d$ than counted arithmetic. Host and CUDA columns average separate retained measurements of all 60 calls; the extra decimals distinguish their means, not timing accuracy.

| $d$ | GFLOPs | Host (ms) | CUDA (ms) | Host block range (ms) |
|---|---|---|---|---|
| 2 | 550.0 | 217.891 | 217.862 | 214.5–220.1 |
| 3 | 561.5 | 269.990 | 269.970 | 265.3–272.6 |
| 5 | 584.4 | 374.321 | 374.300 | 366.8–378.1 |
| 8 | 618.8 | 529.746 | 529.726 | 519.9–536.1 |
| 10 | 641.7 | 629.624 | 629.604 | 623.2–642.2 |
| 20 | 756.2 | 1158.635 | 1158.608 | 1133.5–1172.6 |

tion, or software overhead. CUDA-event intervals are elapsed stream measurements, not summed active-kernel durations. No component-level decomposition is inferred.

### A.2 Resource-coordinate sensitivity

Arithmetic savings and elapsed savings differ. Mean rollout time is 111.9 seconds per attempt for native deployment, 61.8 seconds for $(1, 5)$, and 21.8 seconds for $(2, 2)$. Their mean generated-chunk expenditures are 109.6, 91.7, and 45.0 TFLOPs, respectively. These are observed all-attempt means under the same task allocation; they include different trajectories and outcomes. Table 3 reports all discovery configurations.

There is also an ordering reversal: $(1, 5)$ uses more arithmetic per attempt than $(2, 20)$ (91.7 versus 67.8 TFLOPs), but less rollout time (61.8 versus 107.8 seconds). These means include different trajectories. On a fixed input, reducing $d$ from 10 to 5 lowers mean generation host time from 629.6 to 374.3 ms (40.5%), despite only 8.9% less counted arithmetic. The calibration therefore also separates arithmetic savings from elapsed savings without trajectory differences. It does not resolve their component-level cause. A FLOP-constrained choice is not necessarily a latency- or throughput-optimal choice.

As a descriptive sensitivity analysis, define calibrated host-time rate $b_T(h, d) = \overline{T}_{\mathrm{gen}}(d)/h$, the amortized fixed-input generation time per simulated control step, and normalize it by native host-time rate. For $(1, 2)$, $(2, 2)$, and $(5, 2)$, the normalized arithmetic rates are 0.857, 0.429, and 0.171, while the normalized host-time rates are 0.346, 0.173, and 0.069. Consequently, $(1, 2)$ is ineligible at a half-native arithmetic ceiling but eligible at a separately normalized half-native host-time ceiling. Figure 2(b) shows the resulting choices at all three ceilings. These resource constraints admit different refresh schedules; the choices do not establish quality optima. The frozen experiment remains arithmetic-based.

### A.3 Language-shape transport audit

The profiled batch contains one sample, two $3 \times 360 \times 360$ image tensors, an eight-dimensional state tensor, and language-token and attention-mask tensors of length 20. The checkpoint exposes seven-dimensional actions and generates a 50-action chunk. Evaluation fixes batch size, image preprocessing, state/action dimensions, and chunk size. The tokenizer pads language only to the longest sequence in a batch. Reconstructing the declared task strings with the pinned tokenizer gives 15–17 tokens for discovery and 5–12 for transfer, versus 20 in calibration. We profile every distinct evaluated length for $d \in \{2, 5, 10\}$, preserving all other shapes, and test eligibility against the original absolute ceilings. Table 5 reports the range over evaluated lengths 5–17 against the original length-20 value.

Table 5: Dense generation arithmetic across evaluated language shapes. Budget tests retain the original absolute ceilings; none of the reported transfer configurations changes eligibility.

| $d$ | Evaluated lengths GFLOPs (range) | Length 20 GFLOPs | Eligibility changes |
|---|---|---|---|
| 2 | 539.7–547.9 | 550.0 | 0 |
| 5 | 573.5–582.2 | 584.4 | 0 |
| 10 | 629.8–639.3 | 641.7 | 0 |

In particular, $(2, 10)$ remains within the original half-native ceiling for all evaluated lengths; shorter inputs reduce rather than increase its counted work. The audit changes neither policy execution nor the frozen selections.

### A.4 Primary expenditure and runtime

The primary selected configuration $(2, 2)$ has a higher success count than $(2, 5)$ on seven tasks and ties on one (Table 8). There are 24 paired wins and seven losses. In a descriptive leave-one-task-out sensitivity analysis, the equally weighted difference over the remaining seven tasks stays positive, ranging from 9.3 to 12.1 percentage points. This post-outcome analysis does not extend inference to a new task population. The supplementary paired records retain all state and seed identities needed to reproduce the primary interval.

Table 6: Descriptive primary expenditure differences, selected minus comparator at $h = 2$, including all attempts. Intervals use the paired state bootstrap conditional on the eight tasks and were added after outcomes were observed. Timing intervals summarize sampling variation in the observed records; they do not account for unmonitored runtime interference during priority completion.

| Mean per attempt | $d = 2$ | $d = 5$ | Difference | 95% interval |
|---|---|---|---|---|
| Reference TFLOPs | 41.1 | 48.7 | $-7.6$ | $[-10.6, -4.5]$ |
| Observed rollout (s) | 19.0 | 34.3 | $-15.2$ | $[-17.2, -13.3]$ |
| Control steps | 149.0 | 166.3 | $-17.3$ | $[-27.9, -6.5]$ |

The retained provenance does not characterize numerical precision and attention execution for every model component. Table 7 therefore supports the stated model configuration, but not component- or kernel-level claims.

### A.5 Complete confirmation outcomes

The four amended comparisons are reported in their documented order, using 20,000 paired state-bootstrap resamples and analysis seed 20260913. Table 1 gives their all-attempt resource means. In particular, the $(2, 2)$ minus $(5, 2)$ expenditure difference is $+25.0$ TFLOPs per attempt, with descriptive interval $[+22.8, +27.5]$. This interval describes generated-chunk arithmetic, including the different trajectory lengths; it is not an elapsed-time comparison or evidence of equivalent quality.

Table 7: Runtime and protocol shared by all four confirmation execution phases. Module versions are the loaded versions reported at evaluation time. The supplement retains source digests and distinguishes conflicting installed distribution metadata from these loaded versions.

| Setting | Value |
|---|---|
| Hardware | RTX 4060 Ti, 16 GB; driver 595.84 |
| Runtime | Python 3.12.3; PyTorch 2.7.1+cu126; CUDA 12.6; NumPy 2.2.6; MuJoCo 3.8.1 |
| Evaluation | Batch 1; synchronous simulation; nominal 20 Hz (50 ms control period) |
| Episode horizons | Spatial and Object: 280 control steps; Goal: 300 |
| Precision controls | Autocast and TF32 disabled; deterministic algorithms enabled |
| Attention and reuse | Eager language/action attention with float32 scores; within-generation KV cache enabled |
| Compilation/warm-up | Model compilation disabled; 20 warm-up generations per condition |
| Evaluator provenance | Source revision and full digest retained in supplement |

Table 8: Confirmation successes by task and configuration. Each task row has 20 attempts: ten paired initial states, each with two policy seeds. The total aggregates 160 attempts per configuration. The primary $(2, 2)$ setting exceeds $(2, 5)$ on seven tasks and ties on one.

| | $h = 1$ | | | $h = 2$ (primary pair) | | $h = 5$ | |
|---|---|---|---|---|---|---|---|
| Task | $d = 2$ | $d = 5$ | $d = 10$ | $d = 2$ | $d = 5$ | $d = 2$ | $d = 5$ |
| Goal 5 | 14 | 15 | 14 | 16 | 14 | 18 | 16 |
| Goal 6 | 18 | 17 | 17 | 18 | 15 | 19 | 18 |
| Goal 7 | 20 | 20 | 20 | 19 | 18 | 18 | 18 |
| Goal 8 | 19 | 18 | 18 | 18 | 14 | 18 | 15 |
| Object 5 | 20 | 19 | 19 | 20 | 20 | 20 | 20 |
| Object 6 | 19 | 18 | 19 | 20 | 17 | 18 | 18 |
| Object 7 | 9 | 9 | 10 | 10 | 7 | 7 | 8 |
| Object 8 | 18 | 20 | 18 | 18 | 17 | 20 | 20 |
| Total (/160) | 137 | 136 | 135 | 139 | 122 | 138 | 133 |

Table 9: Amended descriptive comparisons using all 160 attempts per setting. These contrasts were documented after collection began and are not originally prespecified or fully outcome-blind. Intervals are task-conditional, unadjusted, and do not test a quality-preservation margin.

| Setting A | Setting B | A−B (pp) | 95% interval (pp) |
|---|---|---|---|
| $(2, 2)$ | $(1, 10)$ | $+2.5$ | $[-3.8, +8.1]$ |
| $(2, 2)$ | $(5, 2)$ | $+0.6$ | $[-6.3, +6.9]$ |
| $(1, 2)$ | $(2, 2)$ | $-1.3$ | $[-7.5, +5.0]$ |
| $(1, 2)$ | $(1, 10)$ | $+1.3$ | $[-5.0, +7.5]$ |

**Measurement verification.** Evaluation records preserve identities, paired initial states and observations, generation and denoising counts, action sequences, and outcomes. Reconciliation verifies planned coverage and condition pairing across 87,364 discovery actions. The complete confirmation audit covers all 1,120 unique attempts and 171,805 actions across all four collection phases. It reproduces the unchanged primary subset of 320 attempts and 50,451 actions. Two additional completed repeats have identical outcomes and action sequences; they are checked but excluded from the canonical sample. One interrupted attempt contains ten actions and no completed outcome, so it is recorded separately and excluded. Checkpoint, reset, noise, and environment controls further strengthen the internal-consistency evidence; shared upstream errors remain outside this boundary.

A separate outcome audit covers all 1,122 raw completed confirmation records and 172,243 steps. Every generated action matches the action applied to the simulator, and all recorded outcome fields agree. All 171,302 nonsuccess steps have reward 0, and all 941 success steps have reward 1. This reward agreement is not independent validation because LIBERO derives reward and success from the same task-goal evaluation. Source tracing connects that evaluation to the task's BDDL goal predicates. In a separate parity check, one native Spatial episode also reproduces the upstream evaluator's initial state, observation, 70-action sequence, length, and outcome. These checks constrain recording and execution errors, but do not replace model-free replay in a separate, version-matched simulator with direct inspection of terminal states and task predicates.

**Analysis provenance.** The three budget-specific comparisons, identities, and primary analysis were fixed before transfer outcomes. A deadline-driven scheduling amendment changed execution order. Its record states that outcomes were not used for scheduling; 114 attempts per primary setting had already been completed and recorded when it was written. Policy execution settings remained fixed and all 320 primary attempts were included. Subsequent continuation completes the remaining original identities, not an independent replication. Deduplication retains the earliest completed record under the documented phase precedence.

The original comparisons use analysis seed 20260912 in quarter-, half-, then full-native order; a separate replay resets that seed for the submitted primary interval and reproduces its endpoints. The amended contrasts use seed 20260913. Their amendment was written after collection began, when individual outcomes were visible, and is not fully outcome-blind. The simple, empirical, and compute-only baselines were likewise documented after collection began and are interpreted descriptively. Task disaggregation, leave-one-task-out sensitivity, and expenditure intervals were added after primary outcomes were observed; they do not alter the original comparison or stopping condition. No response model is refitted on confirmation outcomes.

**Analysis artifact.** The accompanying supplement contains the primary episode and action records, planned identities, paired state values, calibrations, frozen predictions, runtime provenance, and analysis code. Its bundled verification procedure reproduces the primary counts, interval, task table, and descriptive expenditure analysis. Documentation specifies the selector features, penalized objective, candidate set, and normalization. Checkpoint weights and simulator assets are not bundled. The continuing repository additionally retains a completed confirmation snapshot and verification code that reproduce the full seven-setting resource table, original and amended comparisons, and task-level counts. The submitted supplement remains separate from these post-submission additions.

## B Frozen Selection and Diagnostic Analyses

We compare two logistic response-model families. A compute-only model uses a quadratic in log normalized rate. An allocation model uses $\log h$, $\log(d/10)$, both squared terms, and their interaction. Each family is fitted with ridge penalty 1 or 10. Leave-one-task-out log loss selects the model, which is then refitted on the discovery set. No individual seed repeat is held out while another repeat from the same task remains in training.

The candidate set is $h \in \{1, 2, 3, 5\}$ crossed with $d \in \{2, 3, 5, 8, 10, 20\}$: the 12 discovery configurations and 12 untested configurations. Their calibrated costs determine eligibility; the fitted model selects the highest predicted success under each ceiling. Ties favor lower compute, then lexicographic order.

Table 10: Discovery-based selector choices. The learned choices and comparisons were frozen before transfer; additional selector baselines were documented after collection began. The empirical selector uses the measured grid; other selectors use the declared candidate set. The simple rule exactly matches the learned choices. Selected rate/native is the learned recommendation's arithmetic rate. Frozen comparators and predicted gaps appear in Table 11; the half-native row is primary.

| | Selector choice $(h, d)$ | | | | Learned |
|---|---|---|---|---|---|
| Ceiling/native | Learned | Simple | Empirical | Compute-only | rate/native |
| 0.25 | $(5, 2)$ | $(5, 2)$ | $(5, 2)$ | $(5, 20)$ | 0.171 |
| 0.50 (primary) | $(2, 2)$ | $(2, 2)$ | $(2, 2)$ | $(2, 10)$ | 0.429 |
| 1.00 | $(1, 2)$ | $(1, 2)$ | $(1, 5)$ | $(1, 10)$ | 0.857 |

Table 11: All original frozen denoising tests: selected minus comparator, holding $h$ fixed within each row. Predicted and observed gaps and intervals are percentage points. The half-native row is primary; secondary intervals are descriptive, task-conditional, and unadjusted.

| Ceiling | Setting $(h, d)$ | | Success difference (pp) | | |
|---|---|---|---|---|---|
| / native | Selected | Comparator | Predicted | Observed | 95% interval |
| 0.25 | $(5, 2)$ | $(5, 5)$ | $+6.5$ | $+3.1$ | $[-2.5, +8.8]$ |
| 0.50 (primary) | $(2, 2)$ | $(2, 5)$ | $+4.0$ | $+10.6$ | $[+4.4, +16.9]$ |
| 1.00 | $(1, 2)$ | $(1, 5)$ | $+2.2$ | $+0.6$ | $[-5.0, +6.3]$ |

The simple rule minimizes denoising count, then chooses the shortest feasible prefix. It reproduces every learned recommendation (Table 10); transfer outcomes cannot distinguish these selectors at the tested budgets. The compute-only choices use the original ridge-10 fit. Its quarter- and half-budget choices are absent from the transfer plan, so this evaluation cannot compare those decisions on transfer tasks.

The selected allocation model has log loss 0.7048, compared with 0.7067 for the best compute-only model. This small difference provides no evidence of an added predictive advantage. Under the same mean natural-log binary cross-entropy, constant $p = 0.5$ scores 0.6931 and an intercept-only model fitted separately on each training fold scores 0.7170. The response-model probabilities should therefore not be read as calibrated task-transfer forecasts. The primary test evaluates a recommendation shared with simpler selectors, so it does not establish an additional benefit from model-based ranking.